Hashan Fernando

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Professional Summary

Data Scientist and Researcher with 4+ years of experience applying deep learning (DL), natural language processing (NLP), large language models (LLMs), topological data analysis, and Bayesian statistical modeling to complex problems in healthcare and public health. Highly effective at translating sophisticated technical research into practical, scalable industry solutions, with hands-on expertise spanning transformer architectures, the adversarial robustness of LLMs, and a proven track record of building robust ML pipelines on large-scale clinical data.

Education

Haslam College of Business, University of Tennessee, Knoxville, TN — 2022 – 2026

M.S. Statistics

GPA: 4.00 out of 4.

Postgraduate Institute of Science, University of Peradeniya, Sri Lanka — 2020 – 2022

M.S. Data Science

GPA: 3.75 out of 4.

Faculty of Science, University of Peradeniya, Sri Lanka — 2015 – 2018

B.Sc. Statistics and Operations Research

GPA: 3.06 out of 4.

Experience

Data Science Researcher — Aug 2022 – Apr 2026

Advanced Computing in Health Sciences Section, **Oak Ridge National Laboratory**, Oak Ridge, TN

- Designed a novel algorithm integrating topological data analysis and Bayesian spatial modeling to identify complex spatial patterns in opioid overdose risk across the USA, reducing uncertainty in baseline Bayesian spatial model estimates.
- Built custom multi-channel CNN architectures in PyTorch to classify regional health risk strata across the USA using high-dimensional topological persistence images.

Researcher — Dec 2019 – Feb 2022

Department of Statistics and Computer Science, **University of Peradeniya**, Sri Lanka

- Developed a smart seal and audio detection system for protecting steel cargo from tampering, leveraging IoT sensors for real-time monitoring.

Data Scientist Intern — Jan 2019 – Jul 2019

VizuaMatix, Colombo, Sri Lanka

- Developed a machine learning model to identify non-performing customers in a banking system.
- Developed a route optimization model for food distribution trucks to improve efficiency and reduce operational costs.

Planning Intern — Oct 2018 – Jan 2019

MAS Holdings, Biyagama, Sri Lanka

- Developed multiple VBA applications for planning inventory.

Selected Projects & Competitions

- **1st Place Winner — MOSSAIC Hackathon**
  ORNL, National Cancer Institute & Leidos Biomedical
  Co-led a research team to empirically demonstrate data privacy leakage in cross-silo Federated Learning architectures by deploying gradient-based adversarial attacks against text transformer models.
- **Clinical Document Classification & NLP Clustering**
  COSC-526: Data Mining & Analytics Project
  Developed an end-to-end NLP pipeline on unstructured clinical notes (MIMIC-III), generating contextual embeddings (Doc2Vec, BioBERT, ClinicalBERT) for supervised classification and applying NER, Mapper-based TDA clustering, cosine similarity, and t-SNE/UMAP for medical terminology extraction and visualization.
- **Spatial - TDA**
  Developed a Python package designed for extracting topological information from spatial data.

Publications

- **H. Fernando**, "Unmeasured Spatial Risk in Disease Modeling: A Topological Data Analysis Approach", *Manuscript in preparation*, 2026
- A. Deas, **H. Fernando**, A. Heidi, A. Kapadia, J. Trafton, A. Spannaus, V. Maroulas, "Identifying spatiotemporal patterns in opioid vulnerability: investigating the links between disability, prescription opioids and opioid-related mortality", *BMC Public Health 25*, no. 1 (2025): 1759 [pdf]
- **H. Fernando**, A. Spannaus, A. Kapadia, "A Geospatial Study of Opioid Overdose Risk in Diverse Regions", *Conference Proceedings for International Population Data Linkage Conference*, vol. 9, no. 5, 2024 [pdf]

Awards

- Merit Award for M.S. in Data Science
- Half Colors, Colors Awarding Ceremony 2017, University of Peradeniya, Sri Lanka

Presentations

- Unmeasured Spatial Risk in Disease Modeling: A Topological Data Analysis Approach, UT-Oak Ridge Innovation Institute's Second Annual Research Symposium, Knoxville, TN, Feb 2025
- A Geospatial Study of Opioid Overdose Risk in Washington D.C. Metropolitan Area, Interdisciplinary Association for Population Health Science Conference, St. Louis, MO, Sep 2024
- A Geospatial Study of Opioid Overdose Risk in Diverse Regions, International Population Data Linkage Network Conference, Chicago, IL, Sep 2024

Technologies

Languages: C, Java, Python, R

Technologies: PyTorch, RStudio, MINITAB, SPSS, Weka, Spark, Google Cloud IoT, Arduino